

Sensorimotor Rhythm or Aggregate Power? An Interpretability Audit of EEGPT on Motor Imagery

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Abstract

EEG foundation models match feature engineered baselines on many tasks, yet recent audits report that they lean on the aperiodic $1/f$ background of the signal and the broadband power that comes with it, more than on the specific oscillations that carry task meaning. We audit EEGPT, a foundation model that adds a spatio temporal representation alignment objective to masked reconstruction and is absent from prior audits, on motor imagery from the PhysioNet dataset (imagery versus rest, 35 subjects). EEGPT decodes the task at about 71 percent within subject, on par with feature engineered baselines, yet its representational geometry is dominated by aggregate, largely aperiodic power (Spearman about 0.24 to 0.32, higher than any single band). We find that whether the model looks rhythm specific depends critically on the analysis: raw band power, which conflates periodic and aperiodic activity, makes a non motor control band survive a total power control as strongly as the sensorimotor bands, while sensorimotor averaging inflates apparent mu specificity. A parameterized, spatially controlled analysis (aperiodic-adjusted oscillatory power, per channel, with control bands, a central versus occipital contrast, and a permutation null) leaves only a small and consistent trend toward the sensorimotor mu rhythm: mu is tracked about 0.04 above non motor control bands, central rather than posterior alpha, and above chance, while beta shows no reliable trend. We frame the mu effect as a trend, not an established component, since at this sample size it is detectable but small. The result characterizes EEGPT as power dominated with at most a weak sensorimotor grounding, and it carries a methodological caution for the interpretability audit literature itself.

1. Introduction

Electroencephalography (EEG) foundation models pretrain on large unlabeled corpora and transfer to downstream clinical and brain computer interface (BCI) tasks, often surpassing pipelines built on hand crafted features [1]. Their accuracy is documented, but a separate question governs trust: is a model's competence grounded in the functional neural signal a clinician would recognize, or in a lower level statistic that happens to correlate with the task.

The EEG power spectrum separates into a periodic component, the oscillations such as the sensorimotor mu and beta rhythms that index specific function [7], and an aperiodic $1/f$ component whose slope tracks the balance of excitation and inhibition and is itself a clinical biomarker [5]. For motor imagery, mu and beta event related desynchronization over sensorimotor cortex is a validated motor signal [7], whereas broadband power moves with muscle activity, movement, arousal, and electrode noise. A model that decodes through the rhythm is interpretable and mappable to physiology; one that decodes through aggregate power reaches the same accuracy on a more fragile basis.

Recent audits report that EEG foundation models lean on the aperiodic background and aggregate power more than on specific oscillations [1, 2, 3, 4]. EEGPT is a useful next case: it is absent from these audits and architecturally distinct, adding a spatio temporal representation alignment objective to masked reconstruction, meant to capture consistent, high signal to noise structure [6]. It is a test of whether a deliberate change to the pretraining objective alters what a model encodes. Answering it turns out to require care, because raw band power conflates periodic and aperiodic activity [9] and spatial averaging can inflate apparent rhythm specificity, so the conclusion depends on the analysis. Our contributions are: (i) an audit of EEGPT, a model and architecture not covered by prior work; (ii) the finding that EEGPT is power dominated with only a small, spatially and spectrally controlled trend toward the sensorimotor mu rhythm; and (iii) a methodological demonstration that the specificity verdict flips with analysis choices, resolved with a parameterized, spatially controlled protocol.

2. Related work

Auditing EEG foundation models. A spectral bias study reports that reconstruction based models encode the aperiodic exponent and offset but under represent oscillations [1]. Broader audits map feature families and spatial grounding [2, 3], and physiologically grounded interpretability maps prediction attributions into spectral and source domains [4]. None includes EEGPT.

Parameterizing spectra. The periodic and aperiodic decomposition follows spectral parameterization [5], with practical recommendations for separating oscillations from $1/f$ activity [10]. Raw band power and band ratios conflate the two, so parameterized measures are preferred [9]. We adopt this throughout.

Representational similarity analysis. RSA compares a representation's geometry to reference geometries by correlating dissimilarity matrices, with control models removed by partial correlation [8].

Sensorimotor rhythms. Motor imagery elicits mu and beta desynchronization over sensorimotor cortex [7], the signal that Common Spatial Patterns [11] isolate.

3. Methods

Data and model. We use PhysioNet motor imagery (imagery runs), imagery versus rest, 35 subjects, cleaned conservatively with Autoreject [12]. We audit the frozen EEGPT encoder, using its 512 dimensional embeddings (braindecode weights, mean pooled) as they are used downstream.

The same cleaned trials feed the model and the spectral references.

Decoding. We decode imagery versus rest within subject (five fold) and across subjects, against Common Spatial Patterns [11] and band power baselines.

Representational similarity analysis. For each subject we build a model dissimilarity matrix from the embeddings (one minus correlation over trial pairs) and compare it to spectral references [8]. To avoid the band power conflation [9], we parameterize each trial spectrum into aperiodic (offset, 1/f exponent) and periodic components [5, 10] and build the oscillatory reference from aperiodic-adjusted power (the residual above the fitted 1/f, integrated per band), per channel, for motor bands (mu, beta) and non motor control bands (theta, gamma). We report partial correlations controlling for the task label and for the aperiodic component, a central versus occipital contrast on mu to separate the sensorimotor rhythm from posterior alpha, and a trial shuffling permutation null. As a reference for aggregate power we also correlate the embedding geometry with a total power matrix.

4. Results

4.1 EEGPT decodes the task and its geometry is dominated by aggregate power

EEGPT decodes imagery versus rest at about 71 percent within subject (chance 50), on par with feature engineered baselines, and transfers modestly across subjects. Yet its representational geometry is dominated by aggregate power: it correlates with total signal power at Spearman about 0.24 to 0.32 (depending on the total-power proxy), higher than with any single band (Figure 1a, 1b).

4.2 The specificity verdict depends on the analysis

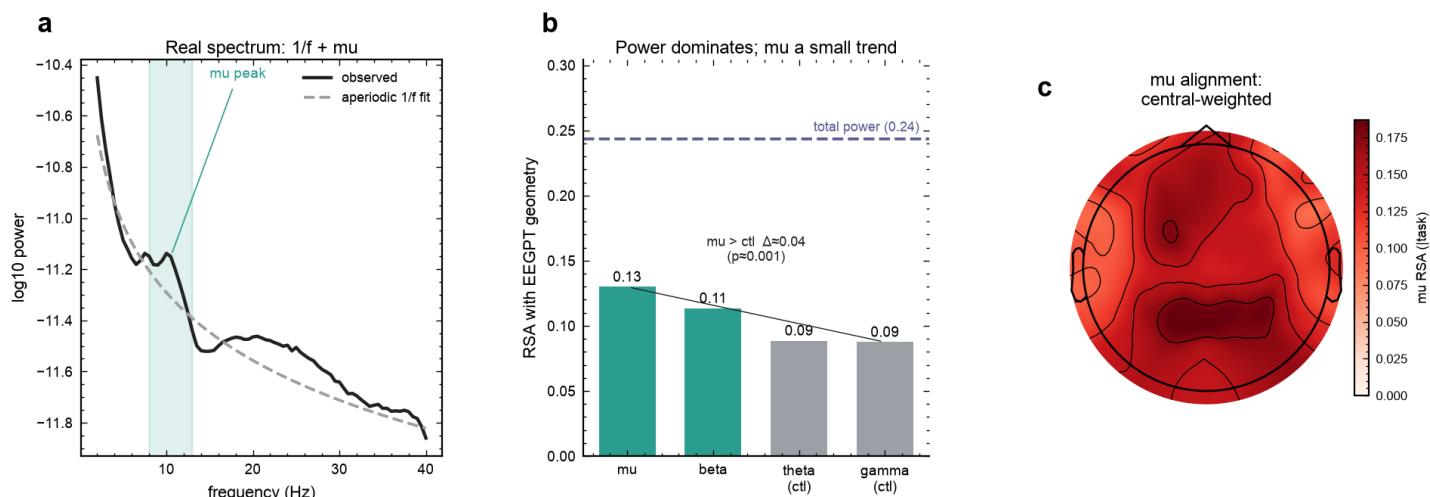
Whether EEGPT looks sensorimotor specific flips with method. Using raw per channel band power with a task and total power double control, a non motor 30 to 40 Hz control band survives as strongly as mu and beta, which would suggest no specificity, but raw band power conflates periodic and aperiodic activity [9], so this test is confounded. Using aperiodic-adjusted oscillatory power on a sensorimotor-averaged spectrum, mu appears strongly specific (correlation about 0.24 versus about 0.05 for controls), but averaging over central channels inflates mu. Neither is decisive on its own.

4.3 A small, controlled mu trend

Holding the spatial construction fixed (aperiodic-adjusted oscillatory power, all 64 channels) resolves it (Table 1, Figure 1c). Motor bands sit only modestly above non motor controls: mu 0.130, beta 0.113, versus theta 0.089 and gamma 0.088. The mu gap is reliable across subjects (mu above theta p approximately 0.002; mu above gamma p approximately 0.0003; positive in roughly 80 percent of subjects), whereas beta is not (p approximately 0.07 to 0.15). The mu effect is central rather than posterior alpha (central mu 0.061 versus occipital 0.027; central survives controlling occipital, p approximately 1e-6) and exceeds a trial shuffling null (whole head mu 0.130 versus null 0.000 plus or minus 0.009, p approximately 0.001). We read this as a small, consistent trend toward a sensorimotor mu component, not an established component: at n = 35 it is statistically detectable but small, and whether it persists and how precisely it can be estimated as sample size grows is open, since significance at a given sample size reflects detectability rather than magnitude.

Table 1. Aperiodic-adjusted oscillatory power tracked by the embedding geometry (per channel, task controlled, n = 35).

band	correlation	motor vs control
mu (8 to 13)	0.130	vs theta +0.041 (p = 0.002); vs gamma +0.042 (p = 0.0003)
beta (13 to 30)	0.113	vs theta +0.025 (p = 0.15); vs gamma +0.026 (p = 0.07)
theta (4 to 7, control)	0.089	reference
gamma (30 to 45, control)	0.088	reference



5. Discussion

EEGPT decodes motor imagery competently, yet its representation is organized primarily by aggregate, largely aperiodic power, with only a small and consistent trend toward the sensorimotor mu rhythm and no reliable beta trend. Despite its added alignment objective, the model does not represent the functional rhythm strongly, which extends the aperiodic and aggregate power bias reported for reconstruction based models to a hybrid architecture. For neurotechnology the practical reading is that EEGPT's competence rests largely on aggregate power, which is not the signal a clinician would rely on and is the part most perturbed when recording conditions shift across sessions or devices.

The audit also carries a methodological caution for the interpretability literature itself: the specificity verdict flipped from non specific to strongly specific to a small trend as we moved from raw band power to sensorimotor averaging to a parameterized, spatially controlled analysis. Band power conflates periodic and aperiodic activity [9], and spatial averaging inflates apparent specificity, so specificity claims about foundation models should rest on parameterized, spatially controlled measures.

Limitations. The audit covers one model, one dataset, and one paradigm; the RSA is correlational; the mu trend is small; the spatial control establishes central weighting rather than full source localization; and we have not run a head-to-head against other foundation models, so we can say the bias persists in EEGPT, not that EEGPT is better or worse than pure reconstruction models.

Future work. A head-to-head RSA against pure reconstruction models (does the alignment objective matter), a cross subject RSA asking whether the transferable component is the rhythm or the power, scaling to more subjects to test whether the mu trend persists, and a causal attribution or erasure test [4] would each strengthen the account.

6. Conclusion

We audit EEGPT, an alignment plus reconstruction EEG foundation model absent from prior audits, on motor imagery. It decodes well but represents the task mainly through aggregate, largely aperiodic power, with only a small, spatially and spectrally controlled trend toward the sensorimotor mu rhythm. The specificity verdict depends on the analysis, which is itself a caution for how these models are audited.

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